Quiz for Clean Kerala :A Statewide Water Quality& Sanitation Challenge

For students of graduation and post-graduation level

Preamble

Water quality and sanitation are among the most pressing concerns of our time. With the increasing incidence of water contamination and related issues, especially in regions like Kerala where traditional water sources such as wells and bore wells are widely used, thereis a growing need for awareness and education on these topics.

This programme is designed with the **objective of creating awareness among the student community about the importance of water quality and sanitation**. Recognizing the role of youth in driving sustainable change, the initiative seeks to equip students with the knowledge and tools necessary to understand and address water-related challenges in their surroundings. We are also prepared to provide study materials and facilitate interactive sessions for students who are interested in taking up assignments or projects related to water quality and sanitation. This awareness programme is a collaborative effort jointly organized by the **Centre for Water Research and Education (CWRE), Unnat Bharath Abhiyan (UBA), Indian Society for Technical Education (ISTE) and National Service Scheme(NSS) Jyothi Engineering College**

The CWRE, functioning under the **Department of Civil Engineering at Jyothi Engineering College**, is actively engaged in research, education, and outreach activities focused on water and sanitation. Our core mission includes **Information, Education, and Communication (IEC)** in the water sector, aimed at empowering communities to address water-related challenges. We operate a government-approved commercial water testing laboratory and offer water testing services at a nominal fee. In addition, CWRE conducts training programmes, publishes books and articles, and undertakes various community-oriented initiatives related to water conservation, quality monitoring, and sanitation awareness. We hope that this initiative will inspire students to become proactive contributors in promoting safe water practices and sustainable sanitation solutions in their communities. We join with UBA, ISTE and NSS to work together to create awareness among the students and general public which we expect to derive advantage

Modality and Eligibility for theWater Quality&Sanitation Awareness Examination

- **Eligibility Criteria:**
  - Students currently studying in degree and post-graduation(B.Tech, M.Tech . B.Sc M. Sc or equivalent
  - Only students from Kerala state are permitted to appear for the examination.
  - Participants must be currently studying in Engineering Colleges/ arts and science college or Polytechnic Colleges and must produce a valid identity proof at the time of the examination.
- **Examination Details:**
  - The exam will be conducted in offline mode at Jyothi Engineering College. Students are required to make their own travel arrangements.

- The question paper will consist of 100multiple choice questions(MCQs).
    - Out of these, questions 61 to 100 are starred and used as tie-breakers if needed.
    - 60 Questions are provided in advance, and minimum 50 Questions of these will appear in the final exam.

- The exam key will be published within 1 day of completion of the examination.
- Opportunity for challenge the key will also be given
- A registration **fee of ₹100** will be charged per candidate.

- **Awards and Prizes:**
    - Attractive cash prizes will be awarded to the top performers:
        - **First Prize: ₹12,000**
        - **Second Prize: ₹8,000**
        - **Third Prize: ₹5,000**
    - All prize winners will also receive Certificates of Merit.
    - All others who attended the exam will get participation certificate

- **Additional Information:**
    - Students may contact the Centre for Water Research and Education (CWRE) for any clarification or assistance.
    - In case of any disputes, the decision of the Administrative Committee will be final.
    - Further details will be made available through the official CWRE website
    - Website - cwre.jeccac.in
    - Date :7th FEBRUARY 2026
    - Time : 11:00 am – 12:00 pm
    - Venue : Jyothi Engineering College, cheruthuruthy, Thrissur.
    - Last date of registration: 31st January 2026
    - Link : https://forms.gle/uTJTbXYR9SJD23FF7
    - REGISTRATION FEE: Rs. 100/-
    - Google pay: cyriac989@okicici

Office bearers for the programme are as follows:

Sl.No	NameandDesignation	Phone number
1	Fr. David Nettikkaden, Executive Manager	
2	Dr.P Sojan lal, Principal	

3	Dr. Alwyn Varghese, HOD CE	
4	Prof. Cyriac .M. G, General Coordinator	9447732619
5	Ms.Anna Joseph, Coordinator	9400947041
6	Mr. Suhas Nair S, Coordinator	6381153948
7	Ms. Jeffy Johny, Coordinator	7025168862
8	Mr.Alfred George, Coordinator	9995890881

(NB:The faculties mentioned in 4,5,6,7,8 may be contacted for any clarifications)

Activity points for participation

Activity Points –KTU -30

For diploma Kerala - 10

Syllabus

Topic: Water Quality and Sanitation Awareness

This syllabus is designed to provide students with a fundamental understanding of water quality, sanitation, and related public health and environmental concerns. The content is aimed at creating awareness and promoting informed action among young learners.

1. Water Quality and Sanitation:An Overview

- Global scenario of water quality and sanitation
- Relevance of water quality in daily life and health
- Apparent effects of water quality issues

2. Water Quality Parameters

- Physical Parameters:
    - pH
    - Turbidity
    - Conductivity
    - Total Dissolved Solids (TDS)
- Chemical Parameters
    - Chloride
    - Alkalinity
    - Hardness
    - Nitrate
    - Sulphate
    - Chemical Oxygen Demand(COD)
    - Biochemical Oxygen Demand(BOD)
- Biological Parameters:
    - Coliform bacteria
    - E.coli
    - Fecal coliform

    - Total coliform
    - Giardia and cryptosporidium

3. Water Pollution

- Heavy metal Pollution in water
- Pesticide pollution and its effects
- Sources and effects of Pollution

4. Water borne and Water-based Diseases

- Common diseases:
    - Cholera
    - Typhoid
    - Leptospirosis
    - Other bacterial and viral infections

5. Water Quality Issues in Kerala State

- Local challenges in water quality
- Community-based and technological remedies

6. Water supply Water Treatment Methods

- Filtration
- Disinfection and Chlorination
- Use of water treatment kits and basic house hold practices
- Domestic water treated systems , RO , hardness ,

7. Sanitation and Waste Management

- Household Sanitation Systems:
    - Septic tank
    - Leach pit
    - Soak pit
- Solid Waste Management at Household Level
    - Sea water intrusion in coastal area
    - Government initiative for sanitation and hygiene

- Role of Local Self Government Departments (LSGD) in managing solid and liquid waste
- Personal and environmental hygiene
- Maintenance of home and surroundings
- Importance of regular hand washing

8. National and Global Initiatives

- Swach Bharat Abhiyan (Clean India Mission)
- Sustainable Development Goals(SDGs)on water and sanitation
    - High tech instruments for water analysis like GCMS , LCMS, ICMS
    - Kerala Solid waste management project
    - Role of green tribal in pollution control

This syllabus form the basis for the multiple-choice questions (MCQs) in the examination. Students are encouraged to explore each topic for both theoretical understanding and practical relevance.

Registration is possible up to 31.01.2026

For any clarification or study support, students may contact the **Centre for Water Research and Education (CWRE).**

Sample questions

1 Drinking water quality standard in India is proclaimed by ….

A) Central Pollution control board **b) BIS**
c) NEERI Nagpur **d) CWPR Oxygen Demand**

2 White deposit on boiling in water indicates presence of

A) Iron **b) hardness**
c) chloride **d) nitrate**

60 questions are given below, in the final exam at least 50 no's of the below given questions will appear as MCQ. Questions 61 to 100 will be starred and used for tie beaker

1. Why is tea prepared with iron-containing water of low quality?
2. Chloride is analyzed using volumetric analysis. Why is blank titration using distilled water performed?
3. The possibility of intrusion of salt water into a well is governed by which principle?
4. Sorenson is popular for which scientific equipment?
5. On what principle does the electrode in a pH meter work?
6. In a well, turbid water is observed immediately after rainfall. What is the possible reason?
7. Why is chlorination to kill bacteria not effective in water containing turbidity?
8. What is the hydrogen ion concentration of a solution with pH = 5.60?
9. How much is the increase in hydrogen ion concentration when pH is reduced from 5 to 4?
10. Why should the presence of Giardia and Cryptosporidium be carefully monitored?
11. Why do authorities encourage planting more trees?
12. Catalox and manganese dioxide are used for removing ______ from water.
13. What is a buffer solution?
14. State True or False: TDS and EC are reasonably comparable.
15. Doxycycline is associated with which disease?
16. Excess nitrate presence in well water represents ______.
17. Jar test has application in ______.

18. Why do we use CO_2-free distilled water for analysis?
19. Work out the quantity of alum and lime in a 6 MLD water treatment plant if the alum–lime ratio is 1:2 and the best lime dose is 9 mg/L.
20. What is meant by chlorine demand?
21. What is the major form of alkalinity and how is it formed?
22. What is excess alkalinity? How do you express it?
23. Why does pH change during aeration?
24. For efficient coagulation, water must be alkaline. Why?
25. Why should wells be provided with sufficient drainage?
26. Organo chlorine is a term associated with ______.
27. Pick the odd one out: a) Purolite b) Catalox c) MnO_2 d) Resin
28. When water is heated, a white deposit is formed. What is your inference?
29. Why does sewage contain more salt than drinking water?
30. Super chlorination is followed by ______.
31. Which award is given by the Indian Government to cities and villages for cleanliness?
32. Acceptable standard for Chemical Oxygen Demand (COD) as per WHO is ______.
33. In a coastal area, the height of water table above sea level is 0.5 m. What will be the fresh water depth below sea water level?
34. State True or False: pH of water after chlorination using chlorine gas was 6.3, but after using bleaching powder it was 6.9.
35. Who is responsible for keeping neighbourhoods clean?
36. What is the full form of ISWM?
37. Churchill and McCay are related to which water quality parameter?
38. On which date does India observe World Toilet Day?
39. Why should soak pits be constructed with limited depth?
40. State True or False: Anaerobic reaction produces methane gas.
41. What are the main objectives of Swachh Bharat Mission (SBM)?
42. How are COD and BOD related?
43. Pick the odd one out: Mercury, Lead, Chromium, Cadmium, Calcium
44. Permissible limit for iron in drinking water as per IS 10500 is ______.
45. Black colour from well water indicates ______.
46. Which method is NOT good for fluoride removal from drinking water: RO / Activated alumina / Bone char / Carbon filter
47. Mass spectrometer is used to measure ______.
48. Pick the odd one out: Biodegradable solid waste, Household liquid waste, Septic tank, Plastic waste
49. What is the reason for lower fluoride presence in water when calcium is present?

50. On heating water, a yellow deposit is observed in a vessel. What is the reason?
51. Black or brownish-black particles in water indicate: a) Iron b) Fluoride c) Nitrate d) Manganese
52. pH of a solution is 2. What is the pOH?
53. State True or False: Acids A and B have the same neutralizing ability, so their pH will be the same.
54. Black tint to water is caused by ______.
55. What is the permissible limit for residual chlorine as per IS 10500?
56. What is the important health significance of hardness?
57. Cholera is a dreadful disease. Which bacteria are responsible?
58. Which agency handles global health issues?
59. Eutrophication in rivers is caused by ______.
60. World Hand Wash Day is celebrated on ______